

Professional SQL Functions Cheat Sheet

This guide summarizes the most commonly used SQL functions in Data Analytics. Each function category includes its purpose and practical use in reporting, dashboards, and interview questions.

Aggregate Functions

COUNT(): Count rows.
SUM(): Total numeric values.
AVG(): Average value.
MIN(): Smallest value.
MAX(): Largest value.

String Functions

UPPER(), LOWER(), LENGTH()/LEN(), CONCAT(), SUBSTRING(), TRIM(), REPLACE(), LEFT(), RIGHT(), REVERSE() are used to clean and format text.

Numeric Functions

ABS(), ROUND(), CEIL(), FLOOR(), MOD(), POWER(), SQRT(), RAND() perform mathematical calculations.

Date & Time

CURRENT_DATE, NOW(), YEAR(), MONTH(), DAY(), HOUR(), MINUTE(), SECOND(), DATEDIFF(), DATEADD() analyze date/time data.

NULL Functions

COALESCE(), IFNULL(), ISNULL(), NULLIF() handle missing values.

Conversion

CAST(), CONVERT() change data types.

Conditional

CASE, IF(), IIF() apply business logic.

Window Functions

ROW_NUMBER(), RANK(), DENSE_RANK(), NTILE(), LAG(), LEAD(), FIRST_VALUE(), LAST_VALUE() support advanced analytics.

Interview Tip: Practice each function with real datasets and combine them with WHERE, GROUP BY, HAVING and JOIN for data analytics projects.